Full Report

"i would like to know more about the problems faced when a patient is unconcious and how old records access can be helpful" *Generated by LeapSpace*

Unconscious Patients and the Value of Old Medical Records at a Glance

Key Findings Table

Key finding	Why it matters in unconscious patients	Population / setting	Evidence
Loss of patient history is the central clinical problem	Clinicians cannot directly confirm symptoms, timing, medications, allergies, past illness, or wishes, which increases diagnostic uncertainty and risk of delayed or missed diagnosis	Emergency and acute-care patients who are unconscious, altered, or critically ill	[1] [2] [3] [4] [5] [6]
Stabilization must often come before diagnosis	Airway, breathing, circulation, oxygenation, and rapid bedside assessment take priority when history is unavailable	ED and critical care	[5] [6] [7]
Old records are most useful for medication and allergy safety	Prior lists reduce omissions, duplications, dose discrepancies, and adverse drug events, though they may still be outdated	ED, trauma, acute care, medication reconciliation studies	[8] [9] [10] [11] [12] [13] [14] [15]
Prior records also help with comorbidity context and handover	More complete records improve documentation of diagnoses, medications, allergies, and care transitions	ED and emergency documentation workflows	[16] [17] [18]
Record access is often incomplete or slow	Missing internal or external data, poor visibility, and paper-chart fallback can delay care and trigger duplicate work	Emergency and hospital informatics settings	[19] [20] [21] [22] [23] [24] [25] [26] [27]

Key finding	Why it matters in unconscious patients	Population / setting	Evidence
Identification is safety-critical before using old records	Wrong-patient orders, mislabeled imaging/specimens, and fragmented identities can make record access dangerous if verification is weak	ED, trauma, CPOE, radiology, registration workflows	[28] [29] [30] [31] [32] [33]
Emergency access is shaped by consent and privacy rules	Proxy, deferred, or presumed consent pathways are used when patients cannot decide, but rules vary and documentation matters	Emergency care, transfer, privacy/consent workflows	[34] [35] [36] [37] [38]

Direct Answer

When a patient is unconscious, the biggest problem is that the clinical team loses the fastest and usually most reliable source of context: the patient's own history. That means staff may not know what happened, when symptoms started, what medicines the patient takes, what allergies they have, what chronic illnesses they live with, or whether there are advance-care preferences. As a result, diagnosis becomes less certain, treatment risk rises, and clinicians often have to stabilize first and investigate in parallel using bedside examination, rapid tests, witnesses, relatives, and any records they can retrieve [1] [4] [5] [6].

Old records can be very helpful because they may reveal medication lists, allergies, chronic diseases, prior imaging, lab trends, and previous notes. The strongest evidence is for medication reconciliation: when prior lists are available and reconciliation is done well, omission, duplication, dose discrepancy, and adverse drug event risk falls, although the records may still be incomplete or outdated [9] [12] [13] [14] [15]. Prior records also support better documentation and handover by giving a more complete picture of diagnoses, medications, and allergies [16] [17].

However, old-record access is not a complete solution. Records may be fragmented across systems, hard to retrieve quickly, hidden inside the EHR, tied to the wrong patient, or constrained by privacy and consent rules. So the best-supported approach is: stabilize the patient immediately, verify identity carefully, and retrieve the most critical prior information as fast as possible from EHRs, shared-record systems, prior hospitals, pharmacies, family, and caregivers [21] [25] [29] [30] [34] [39].

Confidence level: High - The evidence is consistent across multiple emergency, informatics, and medication-safety studies, especially for missing-history problems and medication/allergy benefits of prior records [1] [4] [12] [25].

Study Scope

This report synthesizes evidence from emergency medicine, critical care, hospital informatics, medication reconciliation, patient identification, and privacy/consent workflow studies focused mainly on 2010-2025 acute-care contexts, especially ED and ICU settings [5] [13] [27] [28] [36] [40]. The evidence base is strongest for process issues such as missing history, record retrieval barriers, medication discrepancies, and identification risk, and weaker for unconscious-patient-specific hard outcomes like mortality or time-to-treatment [41] [42] [43].

Assumptions & Limitations

- The literature supports process benefits of old records more strongly than direct outcome improvements specifically in unconscious patients [41] [42] [43] [44].
- Medication lists and allergy records can reduce harm but are not always current, so they cannot be treated as perfectly reliable representations of what is biologically active at the moment of care [14] [15].
- Legal and consent workflows vary by jurisdiction, so no single universal emergency-access model is established across all settings [37] [45].

Suggested Further Research

- Prospective studies specifically comparing unconscious patients with versus without rapid prior-record access on time-to-treatment, duplication of tests, and adverse events [41] [43].
- Comparative studies on which old-record elements have the highest emergency value: medications, allergies, code status, recent imaging, lab trends, or implanted devices [26] [40].
- Better evaluation of identity safeguards such as photos, barcode workflows, and verification dialogs in unidentified or temporary-registration patients [29] [30] [31].
- Implementation studies on how emergency privacy exceptions can be operationalized without slowing urgent care [36] [38] [39].
- Workflow research on “usable interoperability,” not just data sharing, so key old information is surfaced quickly in emergency cognition-friendly formats [21] [27] [39].

Foundational Context

In emergency care, diagnosis usually begins with the patient’s story: what happened, when it started, what the patient feels, what medications they use, what allergies they have, and what their normal baseline is. When a patient is unconscious, that history is absent or unreliable, so the clinical process shifts from history-led reasoning to physiology-led triage and stabilization [1] [2] [3]. In practice, this means teams prioritize immediate threats to life while trying to reconstruct the missing context from examination, tests, witnesses, staff communication, and prior records [5] [6].

A second foundational issue is that “old records” are not a single thing. They may include internal hospital notes, shared longitudinal records, prior medication lists, allergy lists, radiology reports, prior lab values, chronic disease diagnoses, advance-care planning information, and transfer documentation [16] [17] [26] [46]. Their usefulness depends not only on whether they exist, but whether clinicians can retrieve the right information quickly, confirm it belongs to the right patient, and judge whether it is still current [27] [28] [39].

Confidence level: High - These concepts are directly and repeatedly supported across the supplied emergency and informatics literature [1] [6] [17] [27].

Why Unconscious Patients Are Clinically Challenging

Loss of history and diagnostic uncertainty

The main problem in unconscious patients is loss of direct history. Studies describe these patients as harder to diagnose because a good history may be unavailable and the remaining clinical, laboratory, and imaging findings may be nonspecific [4] [5]. This creates uncertainty about the cause of collapse, the timing of events, exposure to toxins or drugs, chronic disease background, and baseline neurologic function [1] [2] [3].

Clinicians therefore have to reconstruct the case indirectly. Evidence describes the use of collateral history from relatives and witnesses, communication with bedside nurses and colleagues, and review of available electronic records to fill information gaps [5] [47] [48]. Taken together, these findings suggest that unconsciousness does not just remove one data source; it disrupts the whole diagnostic workflow by forcing parallel searching and verification under time pressure [5] [49].

Immediate stabilization before definitive diagnosis

Because history is missing and life-threatening causes are possible, stabilization comes first. Evidence emphasizes securing airway, oxygenation, and circulation before pursuing cause-specific treatment [5] [6]. Rapid bedside evaluation is used to identify immediately reversible or catastrophic problems while broader investigation proceeds [7] [50].

This prioritization has an important implication: in unconscious patients, treatment decisions may begin before the diagnosis is fully known [6]. That increases the value of any reliable prior information that can quickly narrow risk, especially drug exposure, allergy status, anticoagulant use, chronic disease burden, and prior neurologic status [12] [26].

Bedside tests and structured assessment

Without history, clinicians lean more heavily on physical examination, CT imaging, laboratory testing, and point-of-care tests such as blood gas, blood count, electrolytes, and lactate [6] [7] [50]. Structured consciousness tools such as GCS or FOUR also standardize assessment, especially when confounders like sedation are documented [5].

There is a trade-off, however. One source warns that untargeted testing can lead to overuse, higher cost, iatrogenic anemia, and unnecessary downstream interventions [50]. This implies that better prior-record access may help not only by adding information, but by making testing more targeted and reducing compensatory over-investigation when history is absent [25] [50].

Confidence level: High - The clinical challenge and compensatory strategies are well supported across multiple emergency assessment sources [4] [5] [6] [50].

How Old Records Improve Safety and Decision-Making

Medication reconciliation and allergy verification

This is the strongest evidence area. When prior medication lists are unavailable or incomplete, admission medication histories are frequently wrong. Reported problems include low accuracy of EMR or triage histories, frequent omissions, and substantial percentages of patients having at least one omitted medication at admission [9] [10] [11]. For an unconscious patient, this risk is magnified because the patient cannot correct the list or identify recent changes [9].

When prior medication lists are available and reconciliation is done effectively, the literature reports fewer omissions, duplications, and dose discrepancies, and some studies report reductions in medication errors and adverse drug events, especially with pharmacist involvement [12] [13] [14] [51]. This makes old records particularly valuable in emergencies where sedatives, anticoagulants, insulin, antiepileptics, or other high-risk medicines may shape immediate treatment decisions [12] [14].

But the evidence also shows an important caution: a documented medication list does not guarantee that clinicians know what is actually present in the patient's bloodstream at that moment [14]. Allergy documentation also does not fully eliminate risk; recorded allergies were still overridden in some cases, even if harm was minor or absent [15]. So prior records improve safety, but they do not replace clinical verification and judgment [14] [15].

Prior diagnoses, chronic disease context, imaging, and lab trends

Evidence for direct outcome effects here is weaker, but the logic is still clinically important. Prior diagnoses and chronic disease context can change how clinicians interpret an emergency presentation, especially when the patient cannot explain their baseline [52] [53]. One out-of-hospital cardiac arrest study found that prehospital documentation of chronic conditions and medications was only moderately accurate compared with hospital records, suggesting that prior history is often incompletely available when the patient cannot speak for themselves [54].

The supplied evidence does not strongly establish that access to prior imaging reports or lab trends specifically improves mortality or time-to-treatment in unconscious patients [41] [42] [44]. Still, taken together, the studies support the inference that prior context helps clinicians distinguish new pathology from longstanding disease and may improve decision coherence, even if direct unconscious-patient-specific outcome proof is limited [52] [53] [54].
Inferred: a compact historical snapshot may be more useful in emergencies than a full undifferentiated chart, particularly for medications, allergies, chronic disease, recent tests, and code status [26] [27].
Confidence level for this inference: Low - It is a synthesis from workflow studies and meta-level patterns rather than direct trial evidence [26] [27].

Documentation quality and handover

More complete records improve emergency documentation of diagnoses, medications, and allergies and improve handover quality [16]. Shared or longitudinal records in the ED are used especially for clinician notes, labs, and radiology, supporting continuity across clinicians and care settings [17] [46]. Emergency documentation systems are also expected to support order/result review and decision-making, suggesting that well-integrated records can reduce preventable error in acute care [17] [18].

This matters in unconscious patients because the initial clinician's reconstruction of the case becomes the foundation for every later handoff. If the prior record helps that reconstruction become more accurate early, downstream care is more likely to be coherent [16] [17].

Confidence level: High - The medication-safety benefit is strongly supported; documentation/handover benefit is moderately to strongly supported [12] [13] [16] [17].

Practical Barriers to Retrieving Old Records

Incomplete EHR access and fragmented information

Even with electronic systems, access to prior records remains incomplete. Studies report missing internal and external documentation, incomplete shared-record visibility, and difficulty finding relevant information quickly at the point of care [19] [20] [21] [22]. Critical information may exist but be poorly surfaced, including advance-care planning details not shown prominently on the main screen [26].

This creates a gap between the theoretical promise of digital records and the practical reality of emergency retrieval [17] [27]. Shared-record access is therefore helpful but often imperfect [17] [46].

Paper-chart fallback and delayed abstraction

Paper records are especially problematic. They are described as difficult to retrieve, time-consuming to abstract, and often incomplete or missing [23] [24]. When electronic access is absent or not timely, this can force duplicate testing and delay decisions [23] [24].

One study reports that incomplete medical data at ED arrival triggered extra phone calls, delays ranging from 5 to 240 minutes, unnecessary procedures, and even avoidable admissions [25]. That finding links record retrieval problems directly to workload and care disruption rather than treating them as mere administrative inconvenience [25] [55].

Workflow, interoperability, and data-quality constraints

Emergency retrieval is hindered by privacy barriers, data-standard mismatch, and data-quality problems when trying to integrate information from multiple sources [39] [56]. Shared ICU and ED systems may depend on network function, may preserve paper-like file structures that hide key details, and are often suboptimal for rapid synthesis of large data volumes during active care [27] [57] [58].

Taken together, these findings suggest that interoperability alone is insufficient; the main practical need is usable interoperability, where key historical information is both reachable and rapidly intelligible during bedside work [26] [27] [39].
Confidence level: High - Multiple sources converge on incompleteness, delay, and usability problems [19] [20] [25] [27] [39].

Identity, Consent, and Emergency Record Access

Patient identification and record linkage

Old records only help if they belong to the right patient. Wrong-patient order entry occurs in ED workflows, and verification interventions reduce these errors. Reported safeguards include verification dialogs, ID re-entry, and photo display in the EHR, with studies showing modest to meaningful reductions in wrong-patient errors [28] [29] [30] [59]. Barcode and armband workflows, education, and computer-assisted positive identification also improved identification behavior and reduced errors [31] [33] [60] [61].

However, no single safeguard is enough. One study found that limiting open charts did not clearly change intercepted wrong-patient medication orders [62]. Radiology near-miss events also showed that misidentification can persist despite dual-identifier rules [32]. This supports a layered-safety approach rather than reliance on a single workflow policy [29] [32] [62].

Consent, proxy decision-making, and emergency exceptions

For incapacitated patients, studies emphasize legal representatives, substitute decision-makers, and presumed or deferred consent pathways when the patient cannot decide [34] [35] [36] [63]. Emergency care law also centers on evaluation, stabilization, and appropriate transfer, with careful documentation required during urgent care transitions [64] [65] [66] [67].

But jurisdictions vary. European emergency research sources note that deferred patient or proxy consent may be allowed but not uniformly, and some jurisdictions report legal uncertainty or gaps [37] [45]. As a result, emergency teams may face tension between the need for speed and the need to justify and document why records were accessed before patient consent could be obtained [36] [38].

Workflow safeguards and documentation

Structured forms, standard operating procedures, and secure transmission methods are proposed to reduce legal and clinical risk during emergency record access and treatment delays [38] [39] [68]. One study specifically notes that when treatment is postponed at family request, documentation should record who made the request, why, the urgency level, when reassessment will occur, and the eventual outcome of consent or refusal [38].

The practical message is that emergency record access should be fast, but also traceable: who accessed what, for what urgent reason, under which authority, and how identity was verified [21] [38] [39].

Confidence level: Medium - Identification evidence is strong, but consent/access rules are less uniform across jurisdictions [28] [30] [37] [45].

Practical Implications

For real-world emergency care, the evidence supports a high-yield sequence:

1. **Stabilize first**: address airway, breathing, circulation, and immediate reversible threats [6].
2. **Verify identity carefully**: use multiple identifiers and available safeguards before relying on prior charts for high-stakes decisions [29] [30] [31].
3. **Retrieve a critical historical snapshot**: medications, allergies, anticoagulants, chronic diseases, recent imaging/labs, implanted devices, and code status are likely the most immediately useful elements [12] [26] [27].
4. **Use collateral sources in parallel**: family, witnesses, prior hospitals, pharmacies, and caregivers can help resolve uncertainty [5] [48].
5. **Document the access rationale and authority** when consent cannot be obtained directly [34] [38].

Confidence level: Medium-High - This workflow is a synthesis strongly grounded in the supplied process evidence, though not validated as a single formal protocol in the sources [6] [12] [29] [38].

Conclusion

Unconscious patients are clinically difficult mainly because they remove direct history at the very moment urgent decisions are needed. That loss increases diagnostic uncertainty, medication and allergy risk, and reliance on stabilization, rapid testing, collateral informants, and record retrieval [1] [4] [6]. Old records are most clearly helpful for medication reconciliation, allergy checking, comorbidity verification, and better documentation and handover [12] [13] [16] [17]. Their value is limited, however, by fragmentation, poor usability, delays, wrong-patient risk, and consent/privacy complexity [19] [25] [28] [37].

Taken together, the evidence suggests that the best emergency strategy is not simply “more record access,” but rapid, verified, workflow-friendly access to the right old information at the right time [21] [27] [39].

Overall confidence level: High - Core findings are consistent across the evidence base, with the main uncertainty lying in the size of outcome benefits specifically for unconscious patients rather than in the existence of the underlying problems and process advantages [41] [43].

References

1. Wertz Hartzell, C. (1989) Sleeping beauty: A case of pickwickian syndrome. *Journal of Emergency Nursing*. https://www.sciencedirect.com/science/article/pii/009917678990010X
2. Stubbs, A.M. (2008) Initial approach to the patient with impaired consciousness.. *Missouri medicine*. https://www.scopus.com/pages/publications/51249107405
3. Ahl, B., Weissenborn, K. (2003) Cardinal symptom "acute disturbance of consciousness". *Intensiv- und Notfallbehandlung*. https://www.scopus.com/pages/publications/0348225371
4. Pisciotta, W., Arina, P., Hofmaenner, D., Singer, M. (2023) Difficult diagnosis in the ICU: making the right call but beware uncertainty and bias. *Anaesthesia*. https://doi.org/10.1111/anae.15897
5. Pham, X., Jeffcote, T., Udy, A. (2025) Disorders of consciousness. *Oh's Intensive Care Manual*. https://doi.org/10.1016/B978-0-443-11611-7.00059-1
6. Azharuddin, S., Ogbebor, O., Shuster, M., (...), Cheema, T. (2023) Toxicological Emergencies. *Critical Care Nursing Quarterly*. https://doi.org/10.1097/CNQ.0000000000000439
7. Völk, S., Koedel, U., Pfister, H.-W., (...), Klein, M. (2019) Impaired consciousness in the emergency department. *European Neurology*. https://doi.org/10.1159/000495363
8. Sharma, A.N., Dvorkin, R., Tucker, V., (...), Rosalia, A. (2012) Medical reconciliation in patients discharged from the emergency department. *Journal of Emergency Medicine*. https://doi.org/10.1016/j.jemermed.2011.05.080
9. Monte, A.A., Anderson, P., Hoppe, J.A., (...), Heard, K.J. (2015) Accuracy of electronic medical record medication reconciliation in emergency department patients. *Journal of Emergency Medicine*. https://doi.org/10.1016/j.jemermed.2014.12.052

10. Vouri, S.M., Marcum, Z.A. (2013) Use of a medication reconciliation tool in an outpatient geriatric clinic. *Journal of the American Pharmacists Association*. https://doi.org/10.1331/JAPhA.2013.13084

11. de Winter, S., Vanbrabant, P., Spriet, I., (...), Willems, L. (2011) A simple tool to improve medication reconciliation at the emergency department. *European Journal of Internal Medicine*. https://doi.org/10.1016/j.ejim.2011.03.010

12. Buckley, Mitchell S., Harinstein, Lisa M., Clark, Kimberly B., (...), Kane-Gill, Sandra L. (2013) Impact of a Clinical Pharmacy Admission Medication Reconciliation Program on Medication Errors in "High-Risk" Patients. *Annals of Pharmacotherapy*. https://doi.org/10.1177/1060028013507428

13. Bordin-Wosk, T., Klubnick, J., O'Toole, J.K., (...), Donato, J. (2025) Handoffs, Care Transitions, and Readmissions. *Medical Clinics of North America*. https://doi.org/10.1016/j.mcna.2025.02.016

14. DeAntonio, Jonathan H., Nguyen, Tammy, Chenault, Gregory, (...), Jayaraman, Sudha (2019) Medications and patient safety in the trauma setting: a systematic review. *World Journal of Emergency Surgery*. https://doi.org/10.1186/s13017-019-0225-6

15. Schneider, Philip J. (2013) Evaluating the impact of telepharmacy. *American Journal of Health-System Pharmacy*. https://doi.org/10.2146/ajhp130138

16. Wohlmann, J., Juhra, C., Warnken, L., (...), Eveslage, M. (2021) Medical Emergency Datasets Can Improve the Comprehensiveness of Handover Information: A Simulation Study. *Telemedicine and e-Health*. https://doi.org/10.1089/tmj.2020.0341

17. Landman, A., Gregory Marchand, E. (2014) Information technology in emergency care. *Emergency Care and the Public's Health*. https://doi.org/10.1002/9781118779750.ch5

18. Davidson, S.J., Zwemer, F.L., Nathanson, L.A., (...), Khan, A.N.G.A. (2004) Where's the beef? The promise and the reality of clinical documentation. *Academic Emergency Medicine*. https://doi.org/10.1197/j.aem.2004.08.004

19. Gibby, G.L., Schwab, W.K. (1998) Availability of records in an outpatient preanesthetic evaluation clinic. *Journal of Clinical Monitoring and Computing*. https://doi.org/10.1023/A:1009971521520

20. Finnell, J.T., Overhage, J.M., McDonald, C.J. (2005) In support of emergency department health information technology.. *AMIA ... Annual Symposium proceedings / AMIA Symposium. AMIA Symposium*. https://www.scopus.com/pages/publications/39049188770

21. Lin, Y.-C., Jeng, A., Huang, C.-I. (2005) Rapid identification and provision of medical records for emergency patients. *Journal on Information Technology in Healthcare*. https://www.scopus.com/pages/publications/29244470464

22. Murray, L., Gopinath, D., Agrawal, M., (...), Karger, D.R. (2021) MedKnowts: Unified Documentation and Information Retrieval for Electronic Health Records. *UIST 2021 - Proceedings of the 34th Annual ACM Symposium on User Interface Software and Technology*. https://doi.org/10.1145/3472749.3474814

23. Ahmad Shahabuddin, F., Wah, K.Y., Buji, R.I., (...), Loo, J.L. (2020) Medical record abstraction in a hospital with paper-based records: Experience of early career psychiatrists in northern Borneo. *BJPsych International*. https://doi.org/10.1192/bji.2019.29

24. Devendran, A., Jayam, R., Sindhuja, P. (2015) Electronic medical records using NFC technology. *ARPN Journal of Engineering and Applied Sciences*. https://www.scopus.com/pages/publications/84927135952

25. Krey, L., Rabea, Z., Krause, O., (...), Klietz, M. (2024) Missing Medical Data in Neurological Emergency Care Compromise Patient Safety and Healthcare Resources. *Journal of Clinical Medicine*. https://doi.org/10.3390/jcm13216344

26. Lakin, J.R., Isaacs, E., Sullivan, E., (...), Sudore, R.L. (2016) Emergency Physicians' Experience with Advance Care Planning Documentation in the Electronic Medical Record: Useful, Needed, and Elusive. *Journal of Palliative Medicine*. https://doi.org/10.1089/jpm.2015.0486

27. Rouby, J.-J., Arbelot, C., Deransy, R., (...), Brisson, H. (2014) Electronic medical record in the intensive care unit : Objectives, conception and expected benefits. *Reanimation*. https://doi.org/10.1007/s13546-015-1065-3

28. Adelman, J.S., Kalkut, G.E., Schechter, C.B., (...), Southern, W.N. (2013) Understanding and preventing wrong-patient electronic orders: A randomized controlled trial. *Journal of the American Medical Informatics Association*. https://doi.org/10.1136/amiajnl-2012-001055

29. Green, R.A., Hripcsak, G., Salmasian, H., (...), Vawdrey, D.K. (2015) Intercepting wrong-patient orders in a computerized provider order entry system. *Annals of Emergency Medicine*. https://doi.org/10.1016/j.annemergmed.2014.11.017

30. Salmasian, H., Blanchfield, B.B., Joyce, K., (...), Landman, A.B. (2020) Association of Display of Patient Photographs in the Electronic Health Record with Wrong-Patient Order Entry Errors. *JAMA Network Open*. https://doi.org/10.1001/jamanetworkopen.2020.19652

31. Spain, D., Crilly, J., Pierce, J., (...), Keijzers, G. (2015) Can a barcode scanner for blood collection improve patient identification integrity in the emergency department? A prospective before-and-after study. *EMA - Emergency Medicine Australasia*. https://doi.org/10.1111/1742-6723.12334

32. Sadigh, G., Loehfelm, T., Applegate, K.E., Tridandapani, S. (2015) Evaluation of near-miss wrong-patient events in radiology reports. *American Journal of Roentgenology*. https://doi.org/10.2214/AJR.14.13339

33. Pablo-Comeche, D., Buitrago-Vera, C., Meneu, R. (2010) Evaluation of the launch of an unambiguous patient identification strategy in the hospitals of the Valencian Healthcare Agency (Spain). *Medicina Clinica*. https://doi.org/10.1016/S0025-7753(10)70021-X

34. Kompanje, E.J.O., Maas, A.I.R., Menon, D.K., Kesecioglu, J. (2014) Medical research in emergency research in the European Union member states: Tensions between theory and practice. *Intensive Care Medicine*. https://doi.org/10.1007/s00134-014-3243-6

35. Chidwick, P., Sibbald, R.W., Hansen, T.-L., Parkes, C. (2013) Managing access and flow through appropriate discharge: preventing common errors and improving processes. *Healthcare quarterly (Toronto, Ont.)*. https://www.scopus.com/pages/publications/84928701434

36. Martinez, J.A. (2012) Legal issues in emergency medicine. *An Introduction to Clinical Emergency Medicine*. https://doi.org/10.1017/CBO9780511852091.056

37. Doherty, A.M., Matthews, P.R. (2025) Challenges in decision-making capacity in the acute hospital setting: two years on from the implementation of the Assisted Decision-Making (Capacity) Act 2015. *Irish Journal of Psychological Medicine*. https://doi.org/10.1017/ipm.2025.10118

38. Surianto, A.T.S., Isrul, M., Sya'ban, A.R., (...), Hadju, L. (2025) Addressing Legal Uncertainty in Delayed Informed Consent: Protecting Health Workers in Indonesian Hospitals. *Volksgeist: Jurnal Ilmu Hukum dan Konstitusi*. https://doi.org/10.24090/volksgeist.v8i2.13089

39. Kawtrakul, A., Chanlekha, H., Issariyakul, T., Khunthong, V. (2017) Cloud-based personal health information broker for emergency medical services. *9th International Conference on Management of Digital EcoSystems, MEDES 2017*. https://doi.org/10.1145/3167020.3167049

40. Fernandes, M., Vieira, S.M., Leite, F., (...), da Costa Sousa, J.M.C. (2020) Clinical Decision Support Systems for Triage in the Emergency Department using Intelligent Systems: a Review. *Artificial Intelligence in Medicine*. https://doi.org/10.1016/j.artmed.2019.101762

41. Zachariasse, Joany M., Seiger, Nienke, Rood, Pleunie P. M., (...), Moll, Henriëtte A. (2017) Validity of the Manchester Triage System in emergency care: A prospective observational study. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0170811

42. https://www.sciencedirect.com/science/article/pii/S1936901820000856

43. Uslu, A., Stausberg, J. (2021) Value of the Electronic Medical Record for Hospital Care: Update from the Literature. *Journal of Medical Internet Research*. https://doi.org/10.2196/26323

44. Sanders, S.F., DeVon, H.A. (2016) Accuracy in ED Triage for Symptoms of Acute Myocardial Infarction. *Journal of Emergency Nursing*. https://doi.org/10.1016/j.jen.2015.12.011

45. Schultz-Swarthfigure, C., Kelly, A.-M. (2021) Is waiver of consent for the use of health information for research acceptable to emergency department patients?. *EMA - Emergency Medicine Australasia*. https://doi.org/10.1111/1742-6723.13728

46. Hripcsak, G., Sengupta, S., Wilcox, A., Green, R.A. (2007) Emergency Department Access to a Longitudinal Medical Record. *Journal of the American Medical Informatics Association*. https://doi.org/10.1197/jamia.M2206

47. Steinkellner, C., Schlommer, C., Dünser, M. (2020) Medical history taking and clinical examination in emergency and intensive care medicine. *Medizinische Klinik - Intensivmedizin und Notfallmedizin*. https://doi.org/10.1007/s00063-020-00731-x

48. Herasevich, S., Pinevich, Y., Lindroth, H.L., (...), Barwise, A.K. (2023) Who needs clinician attention first? A qualitative study of critical care clinicians' needs that enable the prioritization of care for populations of acutely ill patients. *International Journal of Medical Informatics*. https://doi.org/10.1016/j.ijmedinf.2023.105118

49. Daniel, Michelle, Park, SunYoung, Seifert, Colleen M, (...), Mahajan, Prashant (2021) Understanding diagnostic processes in emergency departments: a mixed methods case study protocol. *BMJ Open*. https://doi.org/10.1136/bmjopen-2020-044194

50. Petros, S., Weidhase, L. (2020) Laboratory testing in intensive care medicine. *Medizinische Klinik - Intensivmedizin und Notfallmedizin*. https://doi.org/10.1007/s00063-020-00730-y

51. Phan, Ha, Williams, Macey, McElroy, Kelly, (...), Khandoobhai, Anand (2019) Implementation of a student pharmacist-driven medication history service for ambulatory oncology patients in a large academic medical center. *Journal of Oncology Pharmacy Practice*. https://doi.org/10.1177/1078155219831066

52. Clair, Victoria, Garofalo, Denise M, Wolf, Ariel, (...), Myers, Quintin W O (2025) Diagnosis of chronic disease during admission for emergency general surgery: a portal to healthcare. *Trauma Surgery and Acute Care Open*. https://doi.org/10.1136/tsaco-2024-001645

53. Barfod, Charlotte, Lauritzen, Marlene Mauson Pankoke, Danker, Jakob Klim, (...), Lange, Kai Henrik Wiborg (2012) The formation and design of the 'Acute Admission Database'- a database including a prospective, observational cohort of 6279 patients triaged in the emergency department in a larger Danish hospital. *Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine*. https://doi.org/10.1186/1757-7241-20-29

54. https://www.sciencedirect.com/science/article/pii/S1936901817000110

55. Menon, A.S., Greenwald, S., Ma, T.J., (...), Duriseti, R. (2012) Patient and physician willingness to use personal health records in the emergency department. *Western Journal of Emergency Medicine*. https://doi.org/10.5811/westjem.2011.11.6844

56. Bauer, J., Rohner-Rojas, S., Holderried, M. (2020) Consent management and workflows for cross-sectoral patient records and teleconsultations. *Radiologe*. https://doi.org/10.1007/s00117-020-00655-9

57. Handel, D.A., Wears, R.L., Nathanson, L.A., Pines, J.M. (2011) Using information technology to improve the quality and safety of emergency care. *Academic Emergency Medicine*. https://doi.org/10.1111/j.1553-2712.2011.01070.x

58. Jalilian, L., Khairat, S. (2022) The Next-Generation Electronic Health Record in the ICU: A Focus on User-Technology Interface to Optimize Patient Safety and Quality. *Perspectives in health information management*. https://www.scopus.com/pages/publications/85128555832

59. Reuland, B.D., Redman, C.T., Kneifati-Hayek, J.Z., (...), Adelman, J.S. (2022) Observation and Patients' Perceptions of Incorporating Their Photograph Into the Electronic Health Record. *Journal of Patient Safety*. https://doi.org/10.1097/PTS.0000000000001024

60. Ning, H.-C., Lin, C.-N., Chiu, D.T.-Y., (...), Wu, T.-L. (2016) Reduction in hospital-wide clinical laboratory specimen identification errors following process interventions: A 10-year retrospective observational study. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0160821

61. Henneman, P.L., Fisher, D.L., Henneman, E.A., (...), Nathanson, B.H. (2010) Patient Identification Errors Are Common in a Simulated Setting. *Annals of Emergency Medicine*. https://doi.org/10.1016/j.annemergmed.2009.11.017

62. Kannampallil, T.G., Manning, J.D., Chestek, D.W., (...), Galanter, W.L. (2018) Effect of number of open charts on intercepted wrong-patient medication orders in an emergency department. *Journal of the American Medical Informatics Association*. https://doi.org/10.1093/jamia/ocx099

63. Neitzke, G. (2019) Determining Patients Wishes and Preferences. *Anasthesiologie, Intensivmedizin, Notfallmedizin, Schmerztherapie : AINS*. https://doi.org/10.1055/a-0821-6772

64. Ozcan, Y.A. (2017) Queuing models for hospital emergency departments. *Optimizing Emergency Department Throughput: Operations Management Solutions for Health Care Decision Makers*. https://doi.org/10.1201/b10240-8

65. Rose, M., Newton, C., Boualam, B., (...), Tran, Q.K. (2019) Assessing adequacy of emergency provider documentation among interhospital transferred patients with acute aortic dissection. *World Journal of Emergency Medicine*. https://doi.org/10.5847/wjem.j.1920-8642.2019.02.005

66. Lindor, R.A., Ghaith, S. (2022) Emergency Medical Treatment and Labor Act (EMTALA). *Laws of Medicine: Core Legal Aspects for the Healthcare Professional*. https://doi.org/10.1007/978-3-031-08162-0_6

67. Liu, E.C. (2015) EMTALA: Access to emergency medical care. *Hospital Emergency Departments: Background, Issues, and Trends*. https://www.scopus.com/pages/publications/84955647412

68. Hasavari, S., Song, Y.T., Lawner, B. (2023) Increase Patients' Survivability During Emergency Care Using Blockchain-Based Digital Twin Technology. *Proceedings - 2023 IEEE/ACIS 21st International Conference on Software Engineering Research, Management and Applications, SERA 2023*. https://doi.org/10.1109/SERA57763.2023.10456759